Do field-scale radars blink through the rain?

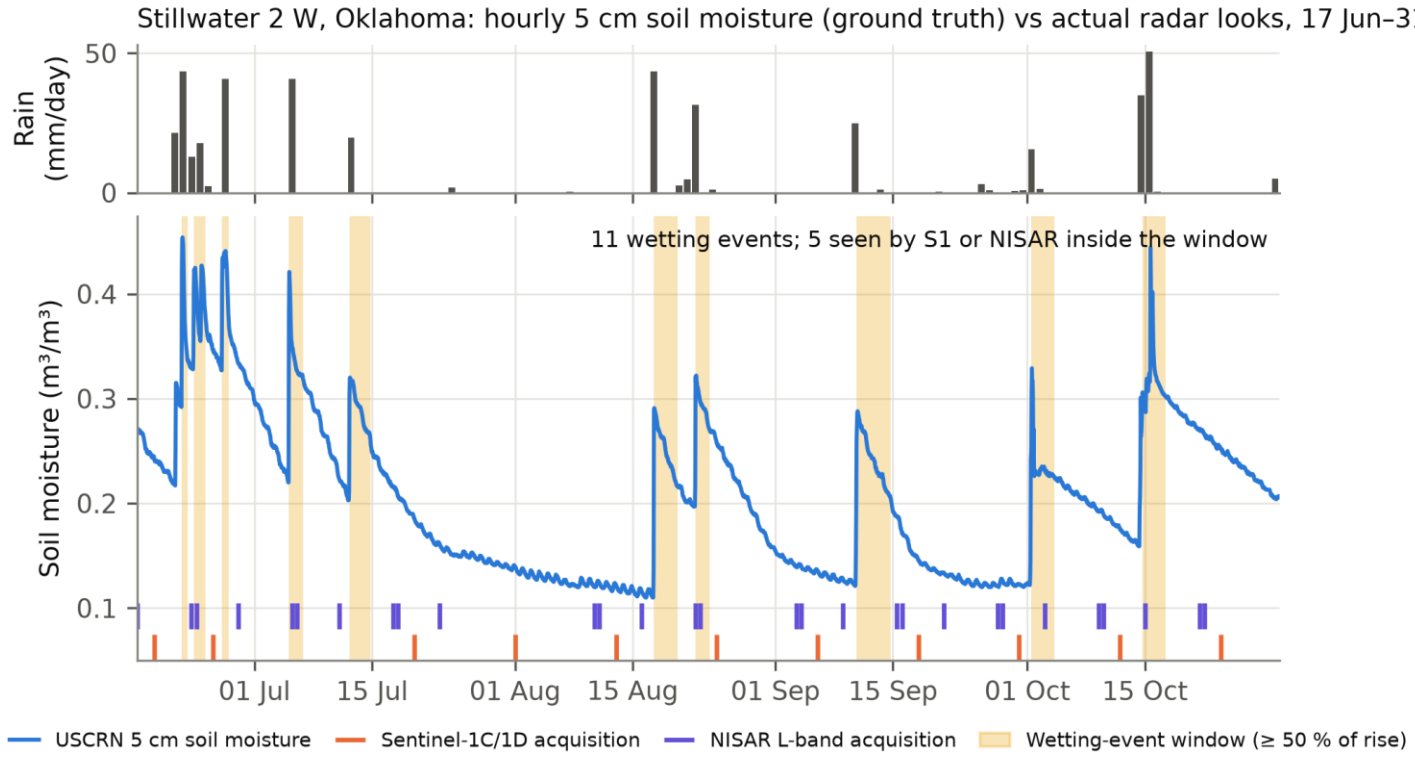
Ground truth from 120 US Climate Reference Network stations tests whether Sentinel-1 and NISAR see soil-moisture wetting events, and makes the case for launching ARMO, a targeted L-band revisit mission

ES4304 Eyes on Earth | NASA ROSES-style proposal | 5 November 2026

Tang Zhen Ian [+ group members]

Every storm writes a two-day signal into the soil

And most satellites look at the wrong moment



- **Moisture jumps within** hours of rain, then decays over 1–3 days
- **That state sets** runoff from the next storm, crop water and land–air fluxes
- **At Stillwater, OK:** 11 events, only 5 seen by any radar look

Resolution or revisit: today you must choose one

Field scale needs SAR, and SAR revisits slowly

40 km

SMAP radiometer

~40 km footprint

2–3 day revisit

6–12 d

Sentinel-1C/1D

10 m C-band SAR

6–12 day revisit; crops attenuate C-band

12 d

NISAR (2025)

200 m L-band soil moisture

12-day repeat; SM ~2× per 12 d

≥ 2030

ROSE-L (planned)

L-band, 6 d over global land (3 d Europe)

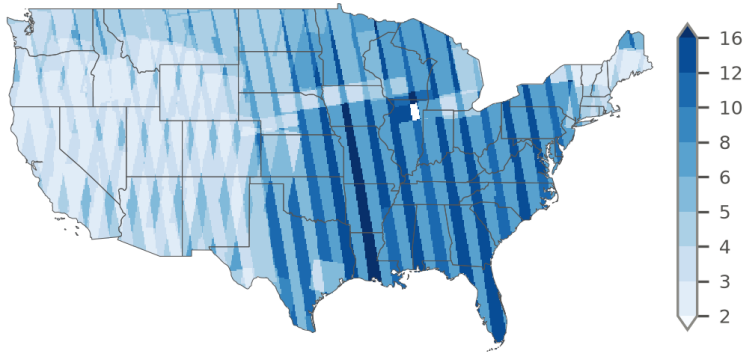
Launch ≥ 2030 (WMO OSCAR)

Physics: L-band (24 cm) penetrates crops and senses ~5 cm; C-band (5.6 cm) senses ~1–2 cm and is attenuated by vegetation.

Where the gap bites: the Midwest and East

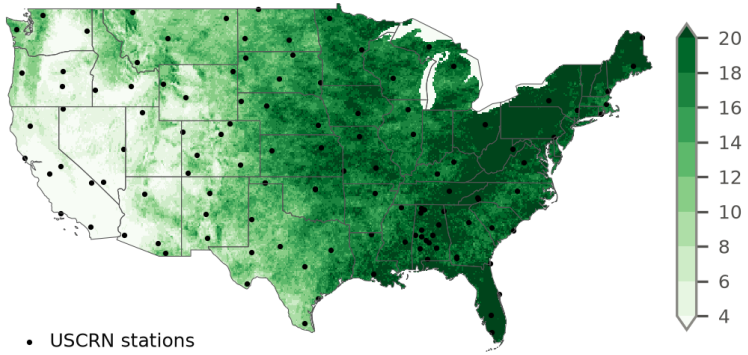
Sentinel-1 revisits slowest where rain, and farms, are most frequent

a Sentinel-1 revisit, 2026 warm season (days)



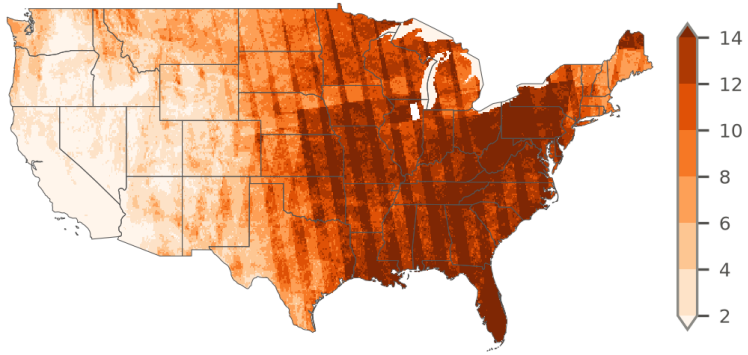
Distinct Sentinel-1 acquisition days, 1 Apr–31 Oct 2026 (Earth Engine COPERNICUS/S1_GRD), as mean revisit. Agrees with our station record (r = 0.90). Median cell 5.6 d.

b Wetting events per warm season (gridMET, USCRN-calibrated)



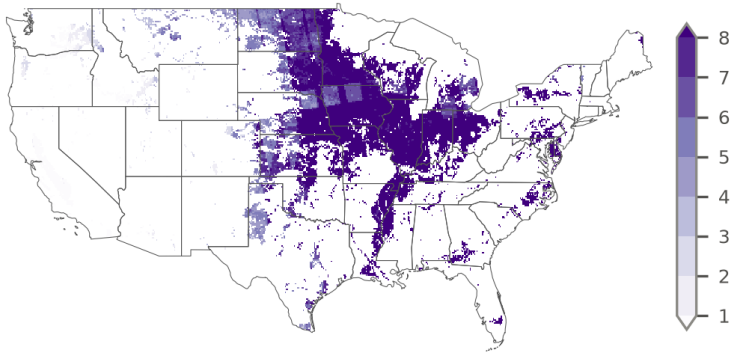
gridMET storm onsets (≥ 5 mm after a drier day), mean of 2025–26, × 0.75 wetting events per onset fitted at USCRN (r = 0.82). Dots: the 120 USCRN stations.

c Wetting events per season missed by Sentinel-1



Events × (1 − capture); capture = E[min(1, W/R)] from the 3,340 USCRN event windows W at each cell's revisit R. Median cell 9 missed per season.

d Extra events per season with ARMO, on cropland (≥ 25 %)

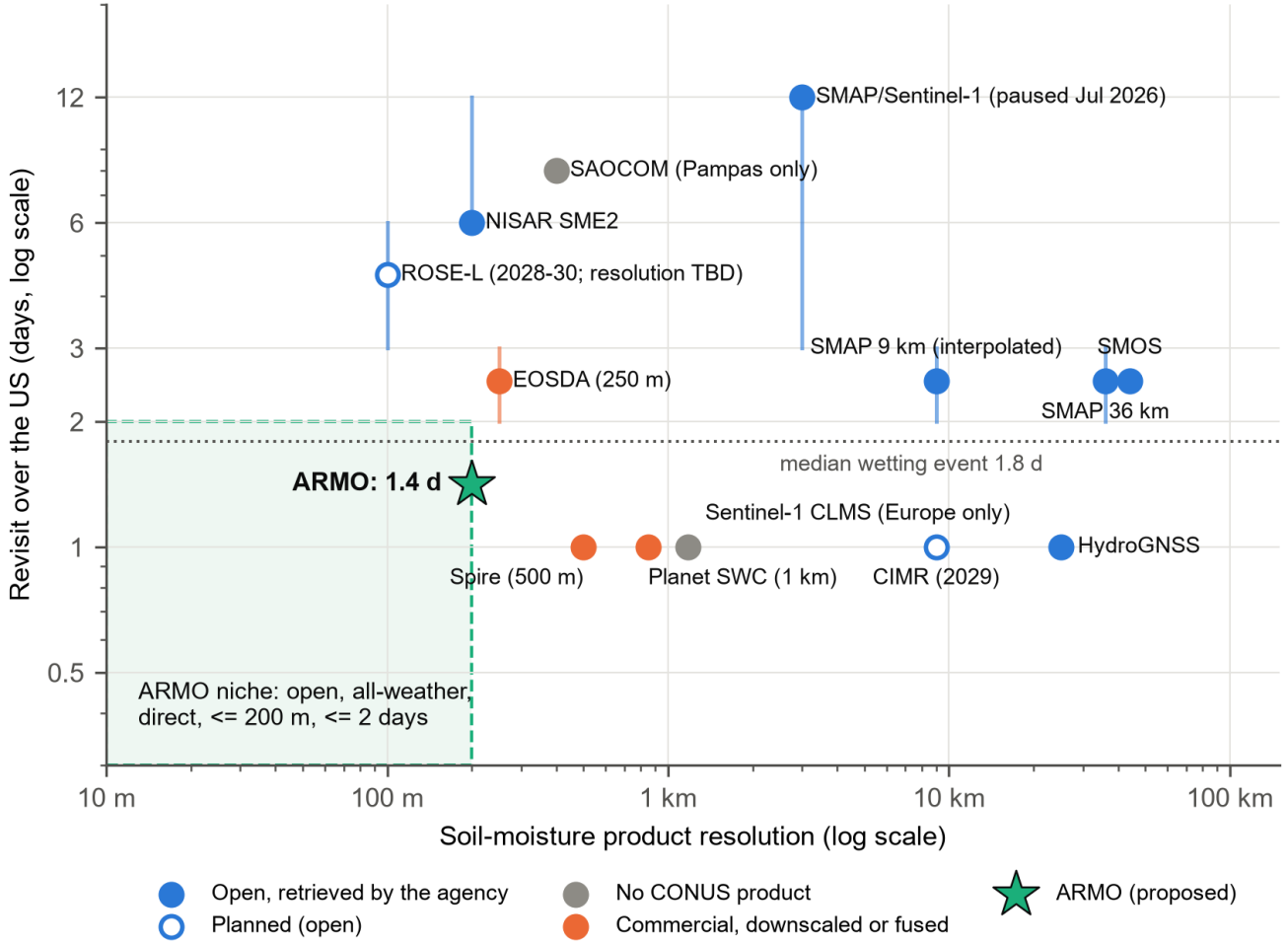


Events gained when a 1.4-day ARMO is added to Sentinel-1, on cells with ≥ 25 % cropland (MODIS MCD12Q1 2024). Median 9 per season.

- **Sentinel-1 revisit:** 5.6 d median, 8–16 d over the Midwest and East
- **Wetting events:** ≈ 15 per season (gridMET, calibrated to USCRN, r = 0.82)
- **Missed by Sentinel-1:** ≈ 9 per season; ≈ 12 on cropland
- **ARMO adds:** ≈ 9 events per season on cropland, mostly the Corn Belt and Delta

ARMO's niche: field scale plus fast revisit

Open, all-weather, ≤ 200 m and ≤ 2 days over the US: nobody fills this box today



- **NISAR SME2:** same 200 m L-band, but ~6-day revisit and 72 h latency
- **ROSE-L:** 6 d global land (ESA); 3 d per eoPortal; launch ≥ 2030
- **Spire, Planet:** daily but modelled from coarse sensors, closed; Planet dropped 100 m
- **Optical:** a clear look in only 56 % of events, even daily
- **Novelty =** revisit, not resolution

Research question

Does today's field-scale radar fleet observe rain-driven soil-moisture wetting events and, if not, what revisit would a dedicated L-band mission need to capture ≥ 80 % of them?

O1 Measure

Event duration and amplitude from hourly ground truth

O2 Quantify

Capture by regular revisits and by actual S1 / NISAR passes

O3 Test

Is a sampled event detectable? S1, NISAR, SMAP, ERA5-Land

O4 Decide

Revisit and noise requirements; marginal value of ARMO

- **H1:** events are shorter than the fleet's revisit, so most are missed
- **H2:** a sampled event's signal is smaller than today's change noise
- **H3:** a dedicated L-band revisit mission lifts capture to ≥ 80 %

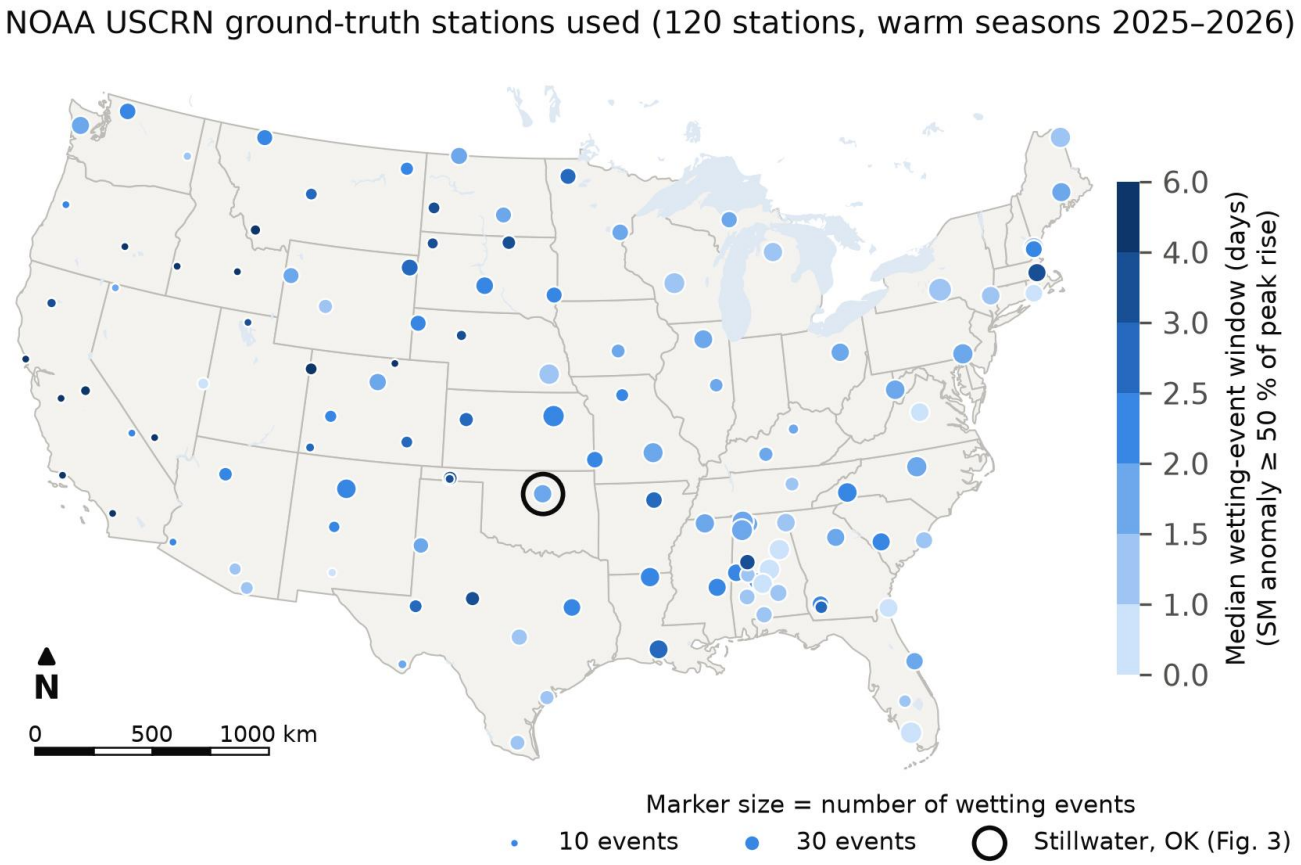
Decadal Survey 2017: question H-4 (floods, droughts)

Soil moisture TO-17: unallocated, no coverage after SMAP

Two stages: retrieval study, then an Earth Venture-class mission

Ground truth first: 120 reference stations, hourly

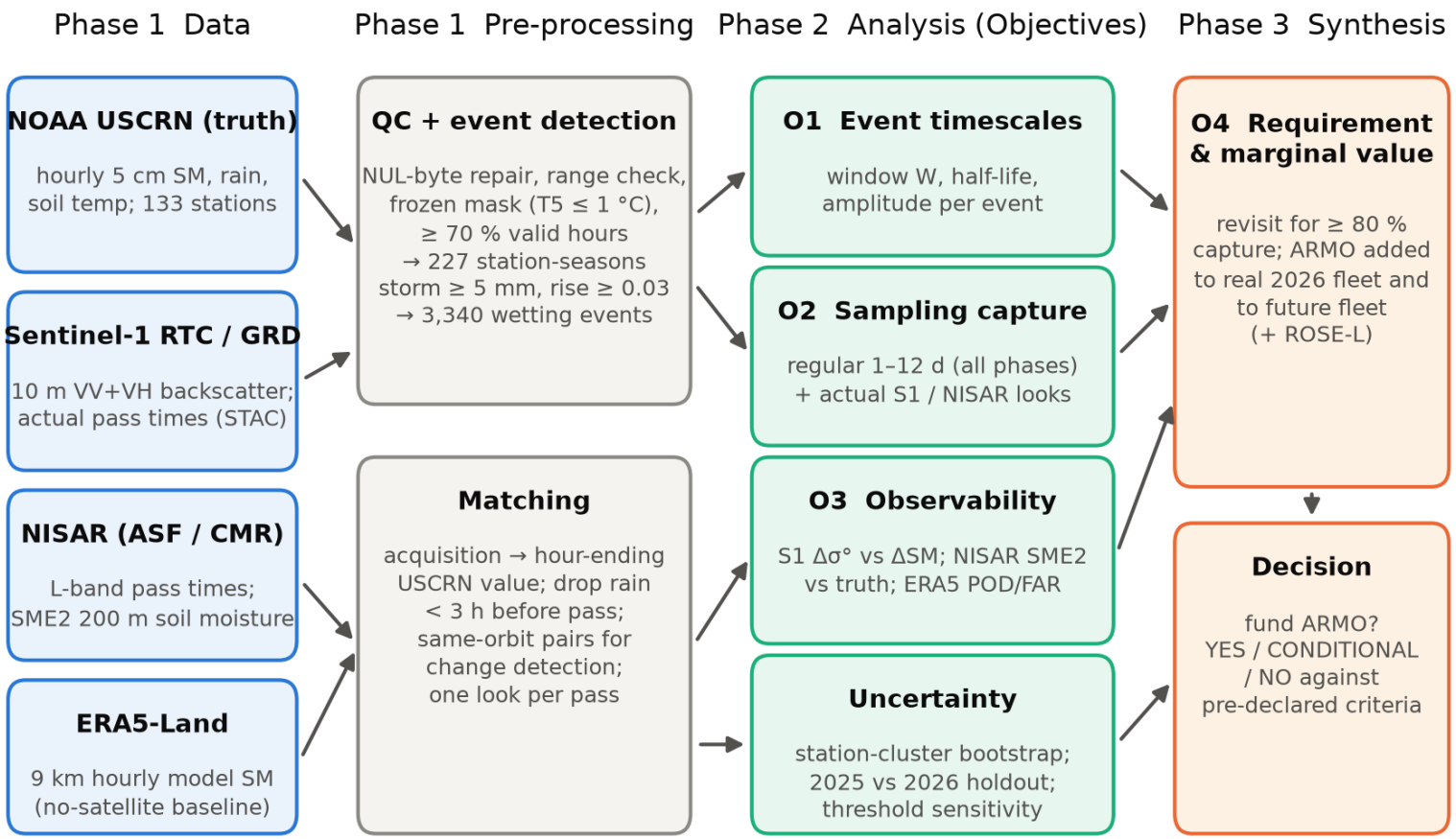
NOAA US Climate Reference Network, warm seasons 2025–2026



Projection: CONUS Albers Equal Area (EPSG:5070). Boundaries: Natural Earth 1:50m.

- **3,340** wetting events at 120 stations
- **Triplicate** 5 cm probes, rain gauge, soil temperature
- **Independent:** used to validate, never to calibrate, satellite products
- **QC:** NUL-byte repair, frozen mask, ≥ 70 % valid hours
- **Land cover (Earth Engine):** 60 grassland, 36 tree, only 9 cropland

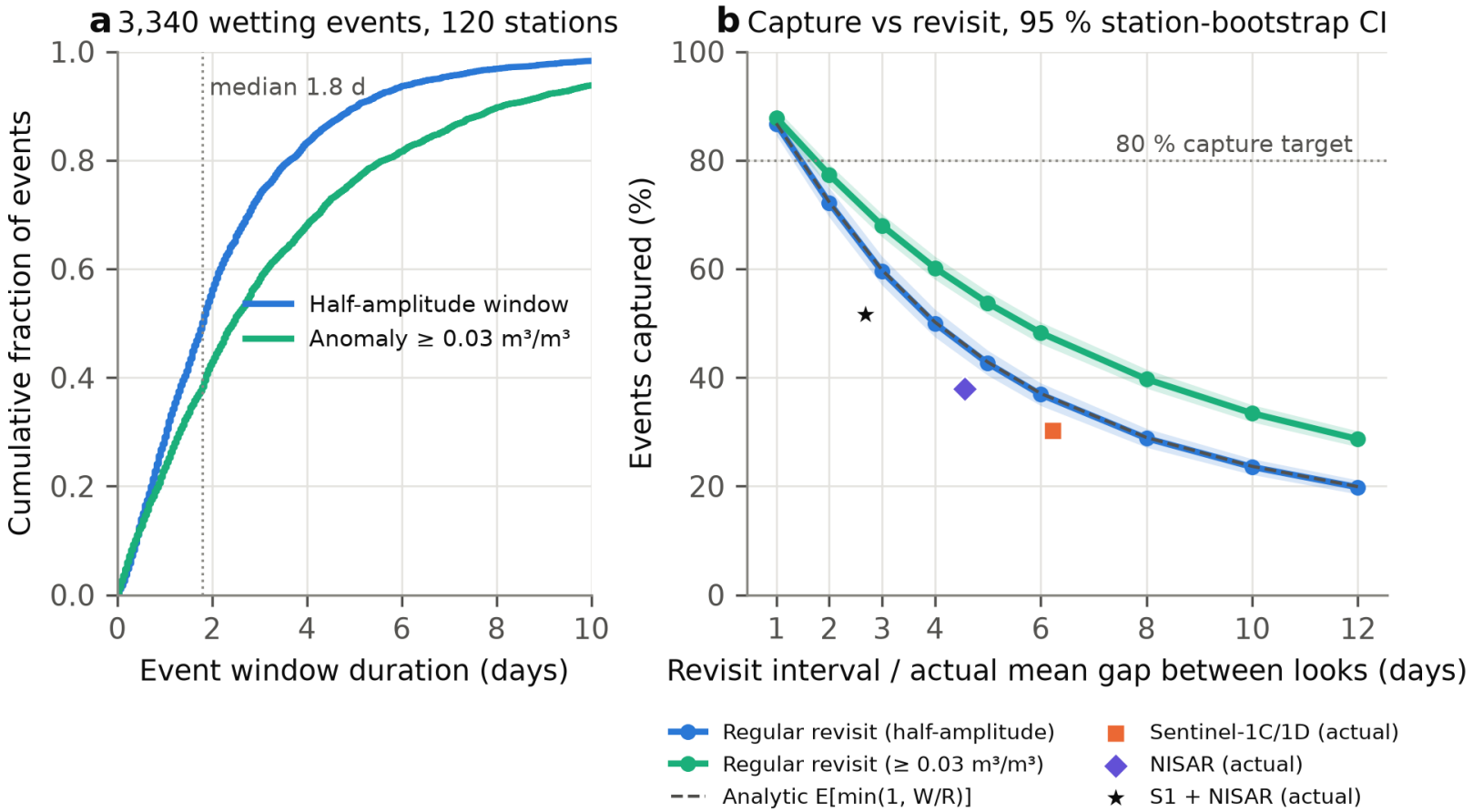
Method: from probe to mission decision



- **Event:** ≥ 5 mm rain and a rise ≥ 0.03 m^3/m^3
- **Window:** time the anomaly stays ≥ 50 % of peak
- **Captured:** a satellite look falls inside the window
- **Real schedules:** 14k S1 and 6k NISAR passes from STAC/CMR
- **Stats:** station bootstrap; 2026 holdout; 9 alternative thresholds

Half of all events fade within 1.8 days

Capture falls quickly with revisit interval



37 %

captured at 6-day revisit

60 % at 3 days; 87 % at 1 day

≈ 1.5 d

for 80 % capture

Regular revisit from one system

The real 2026 fleet sees about half of the events

Even assuming every look were a perfect measurement

30 %

Sentinel-1C/1D

median gap 6.2 d

38 %

NISAR L-band

median gap 4.6 d; max 18 d

52 %

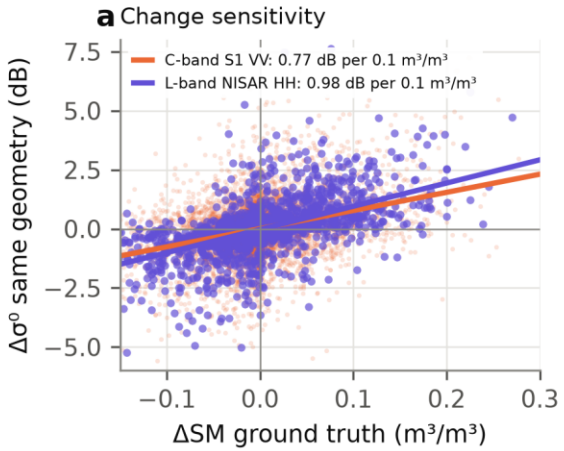
S1 + NISAR

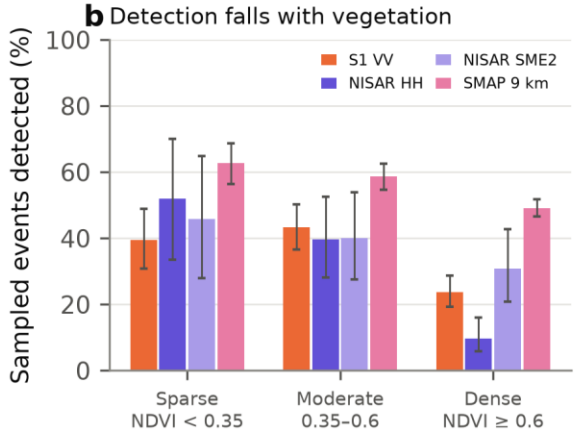
regular sampling at the same rate: 64 %

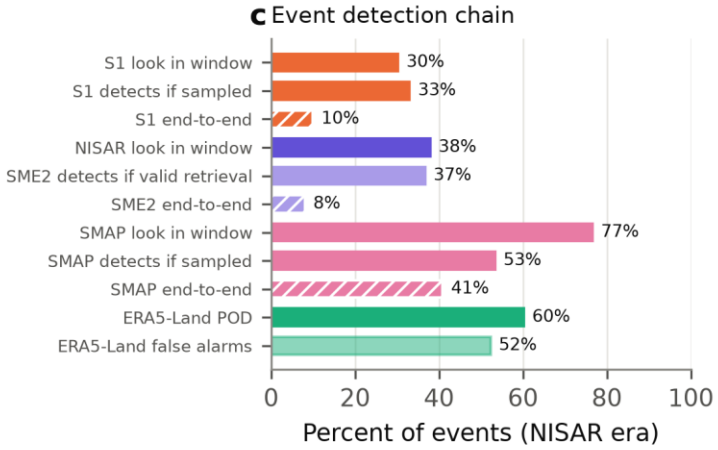
979 wetting events at 107 stations, 17 Jun–31 Oct 2026. Looks cluster on consecutive days, so uncoordinated timing costs ≈ 12 percentage points.

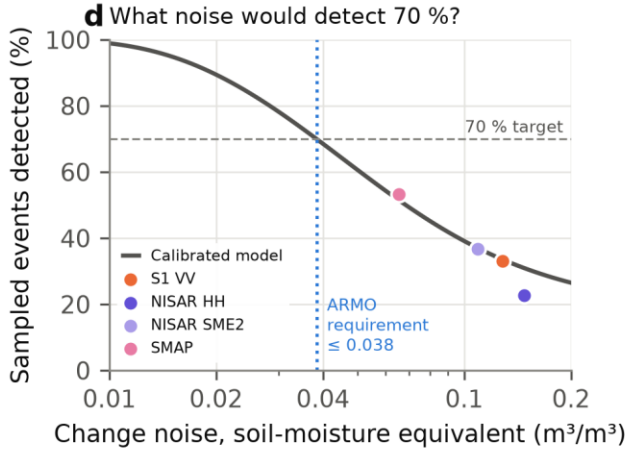
When sampled, can the sensor see it?

Field-scale radars mostly not; the 9 km SMAP radiometer does better









- **Sentinel-1 (C):** detects 33 % of sampled events; noise ≈ 0.13 m³/m³
- **NISAR (L):** 23 % (σ^0HH), 37 % (SME2); noise no lower than C-band
- **SMAP 9 km:** 53 % when sampled, 41 % end to end, but cannot resolve fields
- **ERA5-Land:** POD 60 %, false alarms 52 %

What would it take to see the events?

A detection model calibrated on all four sensors sets ARMO's noise requirement

≤ 0.038

m^3/m^3 change noise

detects 70 % of sampled events (model fits sensors within 3 pts)

0.38 dB

L-band σ^o noise needed

today 1.45 dB: a 3.8× reduction

≈ 15×

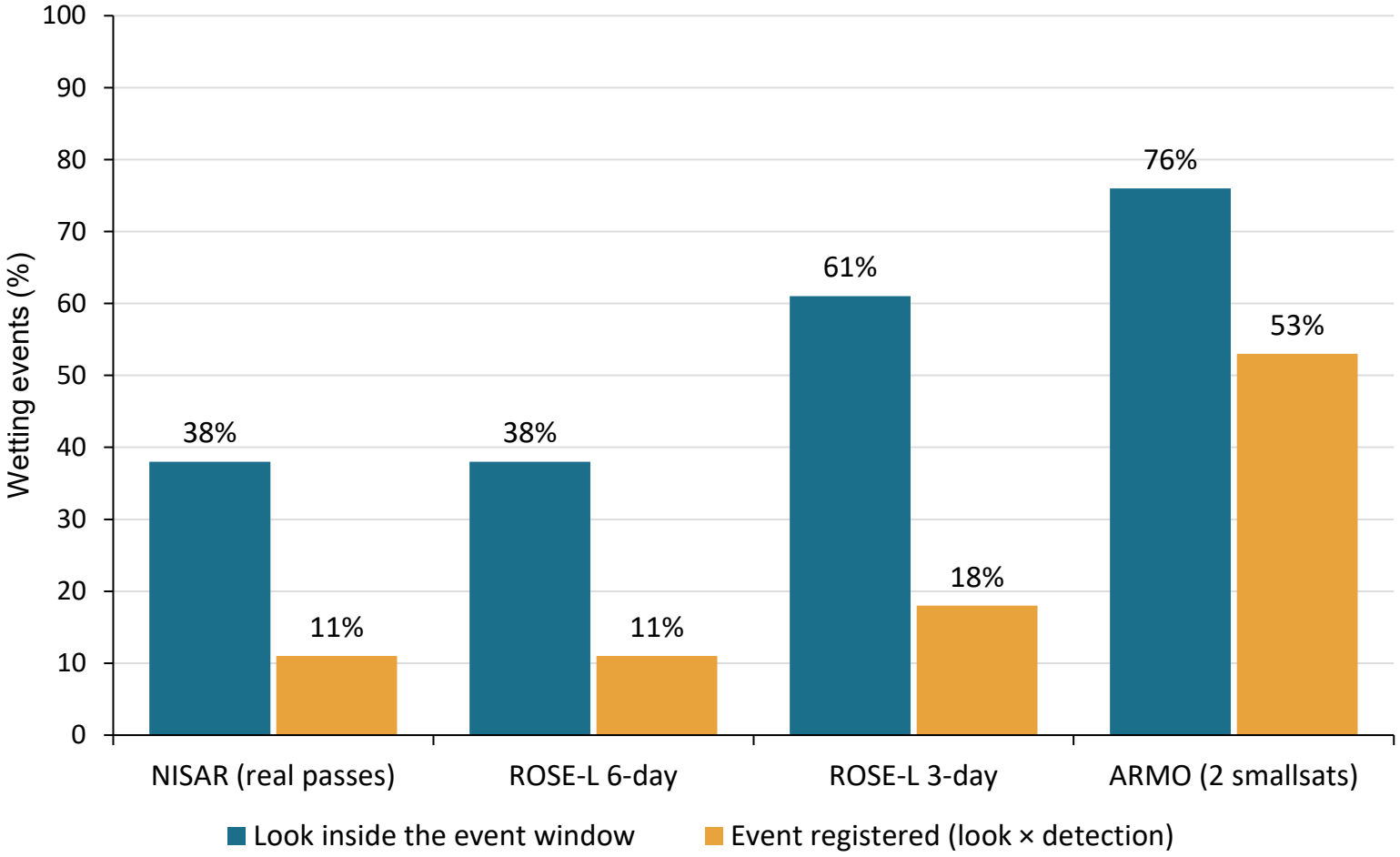
more independent looks

if the noise is speckle: a resolution vs noise trade

This turns the critical risk into a number ARMO's design must meet, and Phase A research must show is reachable (multi-looking, burst averaging, SMAP–SAR fusion).

Head to head: ARMO captures 2× as many events as NISAR

Each system alone, scored on the same 979 ground-truth events (Jun–Oct 2026)



53 %

registered by ARMO, if it meets its requirement

NISAR today: 9–14 %

83 %

with S1 + NISAR + ROSE-L

ARMO lifts capture from 64 % (6-day ROSE-L)

The ARMO mission concept

Two L-band smallsats, one launch, field-scale wetness to forecasters within 24 hours

Fly

2 smallsats, 620 km dawn–dusk orbit, 180° apart

Image

Only the US: 22 min/day, 362 km access, HH + HV

Downlink

45 GB/day to polar ground stations, every orbit

Process

Cloud change-detection retrieval, 100–200 m

Deliver

NOAA National Water Model, USDA, open data, < 24 h

251 kg

per satellite, wet (30 % margin)

5.9 m

mesh reflector, 27 m²

155 W

orbit average; 498 W peak RF

$150 M

incl. launch, 3-yr ops, 30 % reserve

Verdict: retrieval first, then launch ARMO

The revisit gap is real and measured. Find out how good the sensor can be, then launch either way.

Stage 1 (2027–28): retrieval

US$ 218 k, 2 years. Event retrieval on validated NISAR vs USCRN; result sets ARMO's sensor (looks, bandwidth, power)

Stage 2 (2029–36): launch ARMO

Two 251 kg L-band smallsats, one launch ≈ 2033; ≈ US$ 150 M, under the US$ 190 M Earth Venture cap

It pays off either way

Retrieval works: 53 % of events registered, 4–6× NISAR. It doesn't: revisit alone still doubles NISAR (17–28 % vs 9–14 %)

Limitations and the two-stage plan

- **Retrieval:** if 70 % is reached, 53 % of events registered, 4–6× NISAR; if not, still 2×
- **Reflector:** 5.9 m on a 251 kg satellite not yet flown; deployment test before build
- **Cost:** ± 40 %; descope to one satellite (≈ US$ 111 M)
- **Few farms:** only 9 of 121 stations are cropland
- **Scope:** CONUS warm seasons; NISAR still provisional

Stage 1 (2027–28), US$ 0.22 M

Retrieval on validated NISAR vs USCRN; field campaign

Result sets the sensor: looks, bandwidth, power

Stage 2 build (2029–33)

Design, reflector test, build two spacecraft; launch ≈ 2033

Stage 2 operations (2034–36)

Open products to NOAA and USDA in < 24 h

Who would use it, and what it costs

42 %

of US freshwater is for irrigation

USGS Circular 1441 (2015 data)

≈ $30 B

2012 flash-drought damage

NOAA 2013

$218 k

Stage 1: retrieval study now

2 years; personnel-driven; data free

$150 M

Stage 2: ARMO mission

Incl. launch; Earth Venture cap $190 M

- **NOAA National Water Model:** antecedent wetness for flood forecasts
- **USDA Crop-CASMA / FAS:** separate rain-fed from irrigated response
- **Water managers and insurers:** when and how much water reached a field
- **Commercial X-band alternative:** ≈ US$ 19–20 M for the same looks, and physically unsuited

Take-home

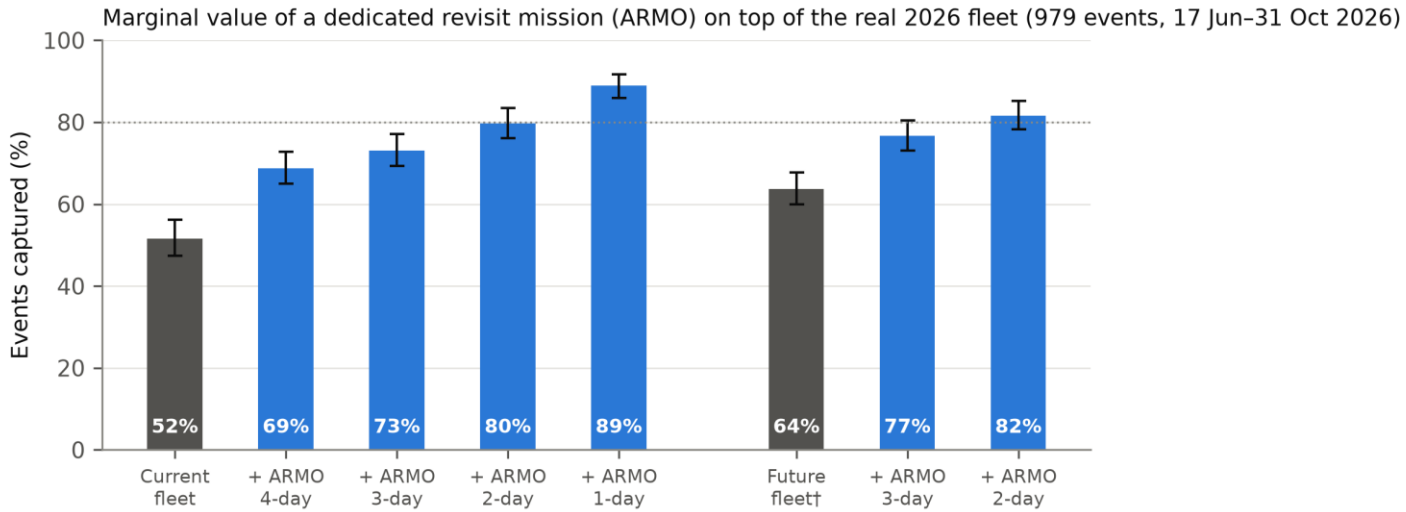
- Wetting events last ~1.8 days; today's fleet captures 52 % at best
- When sampled, field-scale radar detects 23–37 % of events; 9 km SMAP 53 %
- Alone, ARMO captures 76 % vs NISAR 38 % and ROSE-L 38–61 %; with the fleet 81 %
- Stage 1: fund retrieval first (US$ 218 k) to learn how good ARMO's sensor can be
- Stage 2: launch two 251 kg smallsats (≈ US$ 150 M); revisit alone doubles NISAR, retrieval adds the rest
- Questions?

Backup slides

For questions: marginal value, orbit designs, traceability, frequency options

What would ARMO add?

Large gain today; still +18 points once ROSE-L flies



Current fleet = actual Sentinel-1C/1D + NISAR looks. †Future fleet adds ROSE-L at ESA's documented two-satellite revisit over global land (6 days; 3 days is Europe only); launch ≥ 2030 (WMO OSCAR). ARMO = regular dawn/dusk revisit added at random phase. Error bars = 95 % station-bootstrap CI. Dotted line = 80 % capture target.

80 %

with ARMO 2-day, today

from 52 %

+18 pts

added on top of ROSE-L

2-day ARMO, points gained

Can a real orbit deliver it?

Simulated 620 km dawn-dusk orbits, scored against ground truth

Design	ARMO revisit	Fleet + ARMO capture	CONUS imaging duty
1 sat, right-looking, 25-45 deg, dawn-dusk	2.8 d	71 %	1.5 %
1 sat, roll left/right on alternate orbits, dawn-dusk	2.8 d	70 %	1.6 %
1 sat, roll alternate, wider 20-50 deg access	1.8 d	77 %	1.7 %
2 sats (180 deg apart), right-looking, dawn-dusk	**1.4 d**	**81 %**	**1.5 %**
2 sats, roll alternate, dawn-dusk	1.4 d	77 %	1.6 %

Only two satellites clear 80 %. Rolling left/right does not shorten a single satellite's revisit. Regional imaging needs < 2 % of each orbit, far less than global mapping.

ARMO traceability: objective to requirement

Baseline = design D (2 satellites); threshold = design C (1 satellite)

Objective	Requirement	Evidence
Capture ≥ 80 % of events	Revisit ≤ 2 d, with or without ROSE-L	80 % today / 82 % with ROSE-L
Detect ≥ 70 % when sampled	Noise ≤ 0.038 m^3/m^3 (0.38 dB)	Calibrated model; unverified
See under vegetation	L-band ~1.26 GHz, HH + HV	Literature; L-band σ^o ahead only in sparse cover
Field scale	100–200 m, ScanSAR multi-look	Trades against noise
Orbit	620 km dawn–dusk SSO, 180° apart	Orbit simulation (D 81 %, C 77 %)
Spacecraft	2 × 251 kg, 5.9 m reflector, one launch	Sizing: ≈ US$ 150 M incl. launch

Why L-band, not P-band or UHF?

Deeper sensing does not buy longer events, and P-band radar may not operate over the US

P-band SAR (BIOMASS, 435 MHz)

Secondary allocation behind US space-object tracking radars: may not operate over North America

12 m reflector, not a smallsat; 6 MHz bandwidth; ionospheric correction

UHF / P-band signals of opportunity

Receive-only: reuses Navy MUOS signals, no licence needed

≈ 200 m only at swath edge, ~1 km at centre; 7 satellites; one CubeSat demo (SNOOPI)

L-band SAR (ARMO baseline)

Senses ~5 cm, works under crops

Same band as NISAR and ROSE-L: retrievals and calibration transfer

Ground truth at 10 cm (P/UHF depth): 2,531 events instead of 3,340; windows 1.9 d vs 1.8 d; a 2-day revisit captures 73 % (5 cm: 72 %). Way forward: study a UHF receiver as a hosted secondary payload for root-zone moisture.